Unit 9: Introduction to Games Design

Level: **2**

Unit type: **Internal**

Guided learning hours: **60**

Unit in brief

Learners will investigate the features of computer games and how they are designed to meet the expectations of the intended user. Learners will design and promote an idea for a game.

Unit introduction

Computer games development is a large and growing industry that spans the technology and entertainment sectors. From casual games by independent developers that can be played on a smartphone, to the latest 'Triple A' multiplayer game, it is a vast, varied and potentially lucrative industry for computer developers to access.

In this unit, you will explore the basics of computer game design including how the features and design of a game are geared to specific audiences, and the ways in which the expectations of a potential user can be met. You will consider the related technologies and how these affect the gaming experience.

You will apply analytical skills to investigate some existing games and technologies in order to design and promote a game idea in response to a brief.

Learning aims

In this unit you will:

A Investigate how computer games meet user expectations

B Design a computer game in response to a brief

C Present computer game designs to different audiences.

Summary of unit

Learning aim	Key content areas	Assessment approach
A Investigate how computer games meet user expectations	**A1** Features of computer games **A2** Computer game technologies **A3** Target audience	A report exploring the features of computer games for different audiences.
B Design a computer game in response to a brief	**B1** Technical design documentation **B2** Visual design documentation	A portfolio of evidence produced in response to a brief to include: Detailed game designs including a project brief, visual and technical design documents. A prototype of the game. A selection of formal and promotional materials to convey the game idea to technical audiences.
C Present computer game designs to different audiences	**C1** Prototyping **C2** Pitching a game design	

Content

Learning aim A: Investigate how computer games meet user expectations

A1 Features of computer games

Features of games that affect style, gameplay and user experience.

- Genre, e.g. puzzle, simulation, platform, action, RPG, MMORPG, arcade, sports, racing.
- Graphic/visual style, e.g. cartoon, realistic, retro, 2D, 3D.
- Gameplay, e.g. rules, plot, objectives, challenges, goals.
- Game mechanics, e.g. player/computer actions, scoring, win conditions, turns, dice.
- Interaction model, e.g. single player, multiplayer.
- Player perspective, e.g. 1st Person, 3rd Person, world view.
- Input method, e.g. keyboard and mouse, game controller, touch screen, motion controller/sensor.
- Delivery/access method, e.g. digital download/install, online, physical media.

A2 Computer game technologies

Characteristics of different computer game technologies and their impact on user experience.

- Gaming technologies:
    - target devices, e.g. personal computers, consoles, mobile devices
    - web based games i.e. games designed to be played in a browser/ integrated into a web page
    - cloud based gaming services, e.g. Google Stadia®, GeForce NOW®
    - input and output devices, e.g. 3D headsets, game controllers, motion controllers, touch screens, speakers
    - data connections, e.g. Ethernet, Wi-Fi, cellular/mobile networks
    - storage, e.g. hard drive, optical media, cloud
    - new and emerging technologies.
- Benefits and drawbacks of different gaming technologies.
- Considerations and implications for users including:
    - availability
    - costs, e.g. purchasing costs of game, subscriptions, data costs, equipment, in-app purchases

- ease of use and accessibility
- performance, e.g. load times, reaction times, input lag
- user experience, e.g. gameplay, uniqueness, intuitiveness of controls.

A3 Target audience

The characteristics of a game's intended audience and how these affect the design of a game.

- User/audience characteristics including:
  - age
  - interests
  - physical accessibility considerations, e.g. partially sighted, reduced motor function.

- Gaming preferences, needs and experience including:
    - casual or immersive gaming
    - single player or multiplayer
    - play style, e.g. player-vs-player (PVP), player-vs-environment (PVE), co-op (local/online).

Learning aim B: Design a computer game in response to a brief

B1 Technical design documentation

Game design and documentation for technical and logical requirements of a game when responding to a brief.

- Requirements of the brief, including audience and purpose of the game.
- Game features (as listed in A1).
- Game technologies to be used.
- Game logic and structure design including:
    - Storyboard, e.g. storyline, cutscenes, introduction/instruction levels
    - Flowchart, e.g. logic for game mechanics, user input/decisions, game storyline branching
    - Pseudocode, e.g. planning sections of the game code, solving logical problems for game mechanics.

B2 Visual design documentation

Game design and documentation for visual requirements of a game when responding to a brief.

- Game visuals designs including:
    - colour schemes
    - style guide
    - playable and non-playable characters
    - map/level design
    - background music.
- Sources table for assets/content created by others:
    - identify assets/content to be used
    - summary of any editing or manipulation of the asset that will be required,

e.g. cropping, change of file type.

Learning aim C: Present computer game designs to different audiences

C1 Prototyping

Use of game prototypes to demonstrate implementation of select gameplay features.

- Visual style, e.g. playable and non-playable characters, level design, sprite animation.
- Control mechanisms, e.g. player input/control, automated non playable characters.
- Scoring mechanism.
- Asset integration, e.g. raster and vector graphics, animations/videos/ cutscenes, audio.

C2 Pitching a game design

Use and selection of different mediums to present a game design idea to a technical audience.

- Pitching a game to a technical audience including:
    - methods of pitching a game design idea, e.g. formal presentation, business report, feasibility report
    - selection of technical details to be conveyed
    - key features of the game to be included
    - wider considerations, e.g. marketing considerations, trends, costs, uniqueness of idea, similarity to other games
    - repurposing of game content to include in a pitch
    - use of written, verbal and visual assets to convey intended meaning
    - use of tone and language for intended audience and communication method.